

SET A:

1. a) Explain the Knowledge Discovery from Database (KDD) process with its major phases. How does data integration address the challenge of combining data from multiple sources, multiple formats and forms?

b) A numeric variable has the following values. Standardize the values.

Values: 46.93, 138.07, 64.77, 51.46, 15.38, 15.37, 11.32, 223.21, 120.23, 19.85, 19.80, 12.39, 55.79, 86.31, 3.99, 101.51, 30.71, 94.28, 102.39, 18.04

2. a) Differentiate discrete and continuous numeric variables with examples. What are nominal and ordinal categorical variables? Explain with examples.

b) You are given the following dataset of 5 observations with two numeric variables Age and Income:

CustomerID	Age	Income
001	25	50000
002	35	62000
003	30	58000
004	22	48000
005	28	54000

i) Calculate the Pearson correlation coefficient between Age and Income. Show all necessary intermediate steps.

ii) Interpret the strength and direction of the correlation. Support your interpretation with an appropriate plot and briefly explain how the visualization confirms the relationship.

iii) Pearson correlation may not capture non-linear relationships between variables. Why? Explain with an appropriate example.

3. a) The following table presents a hypothetical cross-tabulation between 'food preferences' and 'gender'.

	Female	Male	Other
Likes buff momo	8	75	1
Likes chicken momo	63	3	0
Likes veg momo	23	19	3

Do you think food preferences are correlated with gender? Justify using chi-squared test with significance level 0.05. The critical value for the Chi-Squared statistic with 4 degrees of freedom at a significance level of 0.05 is 9.4877.

b) Explain non-parametric regression and how it differs from linear regression. What is the derivation approach for Nadaraya-Watson kernel regression estimator?

4. a) Describe using matrix formulation how a multivariate linear regression is estimated.

b) The following table shows the values of variables x_1 , x_2 , and y . The fourth column shows the estimated value of y obtained by regressing y on x_1 and x_2 .

x_1	x_2	y	Fitted y
1.0	2.0	8.1	8.0
2.0	1.5	8.4	8.5
1.5	2.5	11.3	11.0
2.5	2.0	11.7	11.5
3.0	1.0	9.2	9.0
1.2	1.8	8.3	8.4
2.2	2.2	12.2	12.0
1.8	2.0	10.0	10.2
2.8	1.5	10.8	10.6
2.0	2.5	12.9	12.5

Calculate the values of R-squared and Adjusted R-squared for the regression model.

5. a) Describe a stationary time series. Why is the stationarity assumption required in time series modeling? Explain.

b) Given the dataset with six observations:

(185, 72), (170, 56), (168, 60), (179, 68), (182, 72), (188, 77),

i) Apply the K-Means clustering algorithm with $K=2$ clusters for two iterations, using the first two points (185,72) and (170,56) as the initial centroids. Show all calculations, including cluster assignments and updated centroids after each iteration.

ii) Interpret the clustering results and discuss whether the algorithm has converged after two iterations. Explain the concept of convergence in K-Means and its significance

6. a) Define autocorrelation and stationarity in time series data. Why are these properties important when fitting ARMA and ARIMA models?

b) How can Principal Component Analysis be considered as a method of identifying latent variables? Explain providing a real-world example.

7. Write short notes on: (Any two)

a) Empirical Distribution

b) AIC and BIC in model selection

c) Permutation Tests for Variable Importance

SET B:

1. a) Explain the difference between structured and unstructured data with examples. Why is data preprocessing important in data analysis?

b) Design a comprehensive strategy for cleaning and transforming a retail dataset that includes customer records with missing values and duplicate entries. The data is stored in heterogeneous formats such as JSON, text files, and Excel sheets, and also includes third-party provider data for user details and behavioral data. The strategy should cover specific techniques for combining heterogeneous data sources, handling missing data, and performing data discretization.

2. a) What is statistical significance? Briefly explain the concepts of p-value and t-value in hypothesis testing.

b) Based on a survey, the variable x takes the following values.

0.76, 1.53, 0.12, 2.44, 2.66, 3.84, 1.03, 6.31, 2.03, 4.61, 9.53, 10.39, 3.69, 6.63,
6.4, 5.01, 2.99, 1.18, 0.35, 0.44, 1.99, 7.04, 3.05, 7.89, 1.44, 1.11, 7.53, 10.4, 1.72, 4.96

Do you think x follows a normal distribution or an exponential-type distribution? Support your answer with appropriate plots.

3. a) What is multicollinearity in regression analysis? How do variance-inflation factors (VIF) help detect it, and what steps can be taken if high multicollinearity is found?.

b) Consider a simple linear regression model: $Y = \beta_0 + \beta_1 X + \epsilon$, where Y is the dependent variable and X is the independent variable. Using the dataset:

Observation	X	Y
1	2	5
2	3	7
3	5	10
4	7	15
5	9	18

- Formulate the matrix representation of the regression model.
- Calculate the Ordinary Least Squares (OLS) estimates β_0 and β_1 .
- Using the estimated regression equation, predict the value of Y when $X=6$.

4. a) Describe the CART algorithm for decision trees and explain how logistic regression is applied in classification problems.

b) You are given the following data.

Observation	X	Y
1	1	-2.3
2	2	0.5
3	3	9.3
4	4	20.8

Assuming that there is an unknown functional relationship $y = f(x)$, estimate the value of y at $x = 2.9$ using the Nadaraya-Watson regression estimator. Take bandwidth $h = 0.5$.

5. a) Differentiate between correlation and causation. Mention examples to justify correlation does not imply causation.

(Unit 5)

b) Given the dataset with six observations:
 (185, 72), (170, 56), (168, 60), (179, 68), (182, 72), (188, 77),

i) Apply the K-Means clustering algorithm with $K=2$ clusters for two iterations, using the first two points (185,72) and (170,56) as the initial centroids. Show all calculations, including cluster assignments and updated centroids after each iteration.

ii) Interpret the clustering results and discuss whether the algorithm has converged after two iterations. Explain the concept of convergence in K-Means and its significance

6. a) Define autocorrelation and stationarity in time series data. Why are these properties important when fitting ARMA and ARIMA models?

b) Apply Principal Component Analysis (PCA) to reduce dimensionality in a customer dataset with 10 features (age, income, spending score, etc.). Explain the mathematical process, interpret the principal components, and determine how many components to retain for 90% variance explained.

7. Write short notes on: (Any two)

a) Zipf's law and its significance in analyzing categorical data distributions.

b) Granger Causality

c) Variable Importance using permutation test

SET C:

1. a) You are given a real-world dataset containing sales data (numeric), product categories (categorical), and promotional periods (time series) from multiple sources.

- Outline the steps you would take to preprocess, analyze, and model this data using techniques from all five units.
- Include discretization and others steps in preprocessing.
- Discuss how you would interpret the results to provide actionable business insights.

b) A numeric variable has the following values.

11.49, 9.59, 12.94, 16.57, 8.30, 8.30, 16.74, 13.30, 6.31, 10.83, 10.47, 10.64, 13.24, 7.09, 7.28, 13.44, 9.99, 8.09, 9.07, 6.30

Discretize the values in ve levels based on the frequencies.

2. a) Below are the values of two covariates x_1 and x_2 .

x_1	0.50	-0.14	0.65	1.52	-0.23	-0.23	1.58	0.77	-0.47	-0.47
x_2	0.25	-0.43	0.94	1.64	-0.37	-0.18	1.36	0.87	-0.60	-0.41

Calculate the Karl Pearson correlation coefficient between x_1 and x_2 .

b) Suppose you surveyed 5 male students and found their average daily spending on refreshments is Rs. 730, with a sample standard deviation of Rs. 200. A previous study shows that female students spend Rs. 810 per day on average. Using a t-test, determine if male students' daily spending is significantly different from that of female students. (Use $|t| > 1.8$ as the threshold for significance.)

3. a) Compare and contrast Principal Component Analysis (PCA) and K-means clustering, explaining their mathematical foundations and practical applications.

b) A real estate company wants to predict house prices (Y) based on area (X_1) and number of rooms (X_2). Given the following data:

House	Area (X_1)	Rooms (X_2)	Price (Y)
1	1200	3	250000
2	1500	4	320000
3	1800	5	380000
4	1000	2	200000
5	1600	4	340000

Calculate the regression coefficients using matrix methods and interpret the R-squared value

4. q) Explain multivariate linear regression with matrix formulation and discuss the problem of multicollinearity using variance inflation factors.

b) A bank wants to predict loan approval (1 = approved, 0 = rejected) based on income and credit score:

Income	Credit Score	Loan Approved
50000	650	1
30000	600	0
70000	750	1
40000	580	0
60000	720	1

Build a decision tree using the CART algorithm and calculate the Gini impurity at each split. Also, set up the logistic regression equation for this problem.

5. a) What is clustering? Explain how K-means can be adapted to handle mixed-type data (numeric and categorical). Describe the different types of variables in data analysis and explain how empirical distributions are analyzed for numeric and categorical data.

b) Explain the difference between $AR(p)$, $MA(q)$, and $ARMA(p,q)$ models. For a time series showing trend, determine the appropriate differencing order (d) and suggest an ARIMA model.

6. a) Discuss the advantages and limitations of using permutation tests for assessing variable importance in machine learning models.

b) Discuss the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) for selecting optimal lag length in time series models.

7. Write short notes on: (Any two) 1×5

a) cross-tabulation

b) Direct Acyclic Graph

c) correlation and causation



MODEL QUESTION SET – 1

Pokhara University

Program: BE – Data Science and Analytics

Time: 3 Hours

Full Marks: 100

1. (a) Explain the Knowledge Discovery in Database (KDD) process with a neat diagram. Discuss each stage in detail. [7]

(b) A numeric variable has the following values:

12.5, 45.6, 78.9, 23.4, 56.7, 89.1, 34.2, 67.8

i. Standardize the data.

ii. Construct a histogram and comment on distribution shape. [8]

2. (a) Explain power-law distribution with real-life examples. Why is it called heavy-tailed distribution? [8]

(b) The following data represents age (X) and annual income (Y in thousands):

X	22	25	30	35	40
Y	25	30	45	50	60

i. Compute Pearson correlation coefficient.

ii. Interpret the result.

iii. Test significance using t-test. [7]

3. (a) What is Ordinary Least Squares (OLS)? Derive the OLS estimator using matrix formulation. [8]

(b) You are given:

Obs	X	Y
1	1	3
2	2	5
3	3	7
4	4	9
5	5	11

- i. Estimate regression coefficients.
- ii. Compute R^2 and Adjusted R^2 .
- iii. Predict Y when $X=6$. [7]

4. (a) Derive the Nadaraya-Watson kernel regression estimator. Discuss bandwidth selection methods. [8]

OR

(b) Given dataset with 3 variables and 5 observations, explain how PCA is computed using eigenvalue decomposition. How do you determine variance explained? [8]

5. (a) At a node of a decision tree, the following data is observed:

A	B	Class
T	T	Yes
T	F	Yes
F	T	No
F	F	No
T	T	Yes
F	F	No

Using Gini index, determine the best attribute for splitting. [8]

(b) Explain CART algorithm for classification trees. [7]

6. (a) Explain ARMA and ARIMA models with mathematical formulation. [7]

OR

(b) How do you select optimal lag length using AIC and BIC? Explain with formula. [7]

(c) What is Granger causality? Why is it not true causality? [7]

7. Write short notes on (Any Two): [2×5=10]

- a) Multicollinearity and VIF
- b) Logistic Regression
- c) Causation vs Correlation



MODEL QUESTION SET – 2

Pokhara University
Program: BE – Data Science and Analytics
Time: 3 Hours
Full Marks: 100

1. (a) Differentiate structured and unstructured data with examples. [7]

(b) Explain data preprocessing techniques:

- i. Data cleaning
- ii. Data integration
- iii. Data transformation
- iv. Discretization [8]

2. (a) Explain empirical distribution for numeric and categorical data. Discuss normal, exponential and binomial distributions. [8]

(b) Given the contingency table:

	Pass	Fail
Male	30	10
Female	25	15

Perform chi-square test of independence. [7]

3. (a) Explain multivariate linear regression model:

- i. Matrix form
- ii. Assumptions
- iii. Multicollinearity problem [8]

(b) Given regression output:

$R^2 = 0.85$, Adjusted $R^2 = 0.80$

- i. Interpret both measures.
- ii. Why is Adjusted R^2 preferred in multivariate models? [7]

4. (a) Explain logistic regression model with derivation of logit function. How are parameters estimated? [8]

OR

(b) Explain K-means clustering algorithm. Discuss distance measures for mixed-type data. [8]

5. (a) Explain permutation-based variable importance and partial dependence plots. [8]

(b) Perform one step of K-means clustering for the following data with $k=2$:

Points: (1,2), (2,1), (8,9), (9,8)

Initial centroids: (1,2) and (8,9). [7]

6. (a) Define stationarity and autocorrelation. How do you test for stationarity? [7]

OR

(b) Explain ARIMA(p,d,q) model. How are p and q selected? [7]

(c) Explain Causal Directed Acyclic Graph (DAG) with example. [6]

7. Write short notes on (Any Two): [2×5=10]

- a) PCA and dimensionality reduction
- b) Statistical significance (p-value, t-value)
- c) Cross-tabulation

Both model sets include:

- KDD process
- Data preprocessing
- Distributions (normal, exponential, power-law, binomial, Zipf)
- Correlation + significance tests
- OLS (matrix derivation + numericals)
- R^2 and Adjusted R^2
- Multicollinearity + VIF
- Kernel regression
- PCA (derivation + interpretation)
- Decision trees + CART + Gini
- Logistic regression
- Variable importance
- K-means clustering
- ARMA/ARIMA + AIC/BIC
- Granger causality
- DAG
- Short notes section

MODEL QUESTION PAPER – SET III

Pokhara University

Level: Bachelor (BE)

Course: Data Science and Analytics (New)

Time: 3 Hours

Full Marks: 100

Attempt all questions. Figures in the margin indicate full marks.

1.a) Explain the Knowledge Discovery in Database (KDD) process with its major phases. (7)

b) A numeric variable has the following values:

12.4, 25.6, 31.2, 45.8, 28.9, 39.5, 41.3, 22.7, 18.6, 35.4

Standardize the values using z-score transformation. (8)

2.

a) Explain empirical distribution. Compare normal, exponential, and power-law distributions with suitable examples. (8)

b) Define Pearson correlation.

Given the following dataset:

X	10	20	30	40	50
Y	15	25	35	45	60

Calculate the Pearson correlation coefficient and interpret the result. (7)

3.a) What is Ordinary Least Squares (OLS)? Formulate multivariate linear regression using matrix notation and derive the OLS estimator. (8)

b) You are given the following data:

x	0.5	1.0	1.5	2.0
y	1.2	2.3	3.1	3.8

Estimate the value of y at x = 1.2 using the Nadaraya-Watson estimator.
Take bandwidth h = 0.3 and Gaussian kernel. (7)

OR

Given a dataset with n variables, explain how Principal Component Analysis (PCA) is computed.
How is the variance explained by each component determined? (7)

4.a) Consider a simple linear regression model:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Given:

X	1	2	3	4	5
Y	3	5	7	9	11

- Write the matrix form of the regression model.
- Estimate β_0 and β_1 .
- Predict Y when X = 6. (8)

b) Explain R^2 and Adjusted R^2 in model evaluation. Discuss multicollinearity and Variance Inflation Factor (VIF). (7)

5.

a) At a node of a decision tree, the following data is observed:

A	B	Class
T	T	Yes
T	F	Yes
F	T	No
F	F	No
T	T	Yes
F	T	No

Which attribute (A or B) should be chosen for splitting using Gini impurity? Show calculations. (8)

b) Describe the CART algorithm. How are splits selected in classification trees? (7)

6.a) What is time series analysis? Explain autocorrelation and stationarity in detail. (7)

OR

Explain ARIMA (p,d,q) model and how optimal lag length is selected using AIC and BIC. (7)

b) What is Granger causality? Explain why Granger causality is not considered true causation. (8)

7. Write short notes on (Any TWO): ($2 \times 5 = 10$)

- a) Data cleaning and integration
- b) Logistic regression
- c) Causation vs correlation



MODEL QUESTION PAPER – SET IV

Pokhara University

Level: Bachelor (BE)

Course: Data Science and Analytics (New)

Time: 3 Hours

Full Marks: 100

Attempt all questions.

1.a) Explain structured and unstructured data with examples. (7)

b) A variable has values:

5, 7, 9, 12, 15, 18, 21, 25, 30

Construct a histogram and comment whether the distribution resembles normal, exponential, or power law. (8)

2.a) Explain numeric and categorical variables.

Describe bar plots, binomial distribution, and Zipf's law. (8)

b) The following contingency table is given:

	Pass	Fail
Male	40	10
Female	30	20

Perform chi-square test of independence at 5% significance level. Interpret the result. (7)

3.a) Derive the Nadaraya-Watson kernel regression estimator. Discuss bandwidth selection rules. (8)

b) Explain Principal Component Analysis (PCA). How are principal components interpreted? (7)

OR

Explain matrix formulation of multivariate linear regression and derive the OLS estimator. (7)

4.a) Consider logistic regression:

Given dataset:

X	1	2	3	4	5
Y	0	0	0	1	1

- i) Write logistic regression model.
- ii) Explain interpretation of coefficients.
- iii) How is it estimated using MLE? (8)

b) Explain permutation importance and partial dependence plots. (7)

5.a) Explain K-means clustering algorithm with steps. Discuss distance metrics for mixed-type data. (8)

b) Explain CART algorithm and pruning methods. (7)

6.a) Explain ARMA model. How does it differ from ARIMA? (7)

OR

Explain autocorrelation function (ACF) and partial autocorrelation function (PACF). (7)

b) Explain causation vs correlation using examples.

Discuss Directed Acyclic Graph (DAG) overview. (8)

7. Write short notes on (Any TWO): ($2 \times 5 = 10$)

- a) Statistical significance (p-value, t-value)
- b) R-squared vs Adjusted R-squared
- c) Power-law distribution